

Computer Graphics – October 9th, 2025

Name: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (24 pages on 12 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- No other material except a calculator is allowed!

I have read and understood the instructions above.

Erlangen, October 9th, 2025 _____

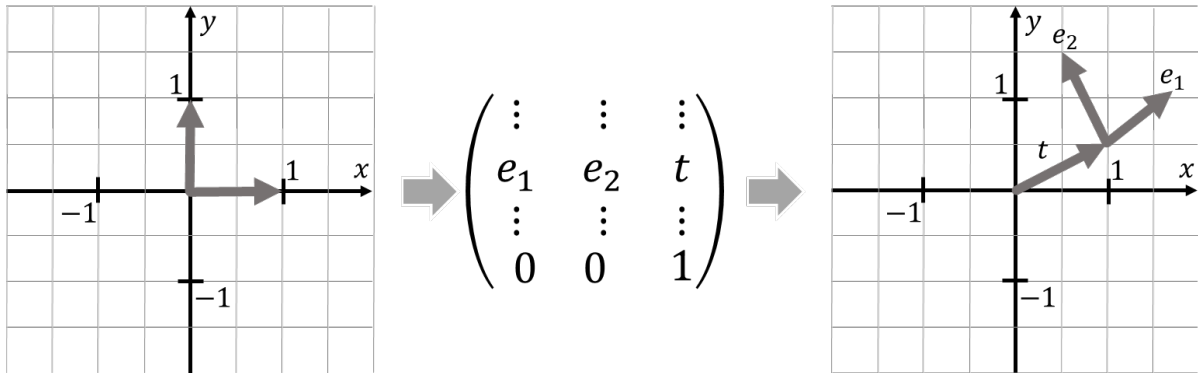
Signature

1	2	3	4	5	6	ALL
12	9	10	11	12	6	60

Grade:

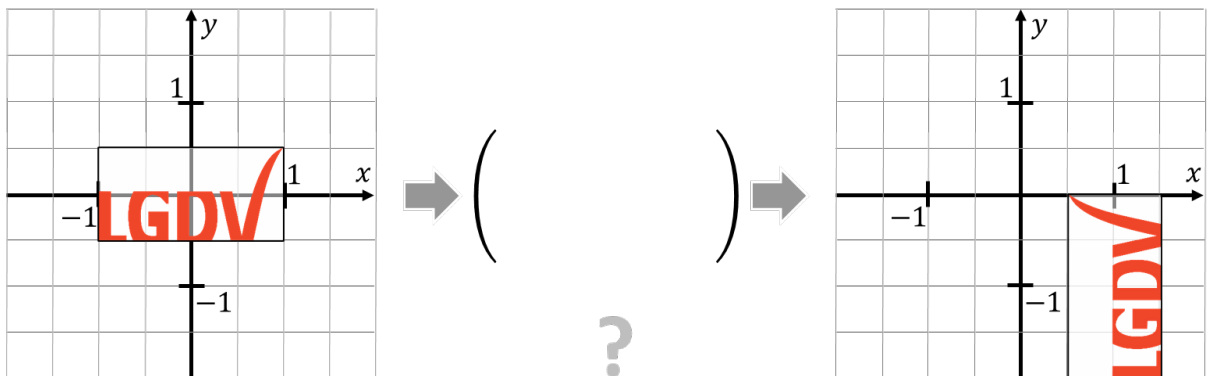
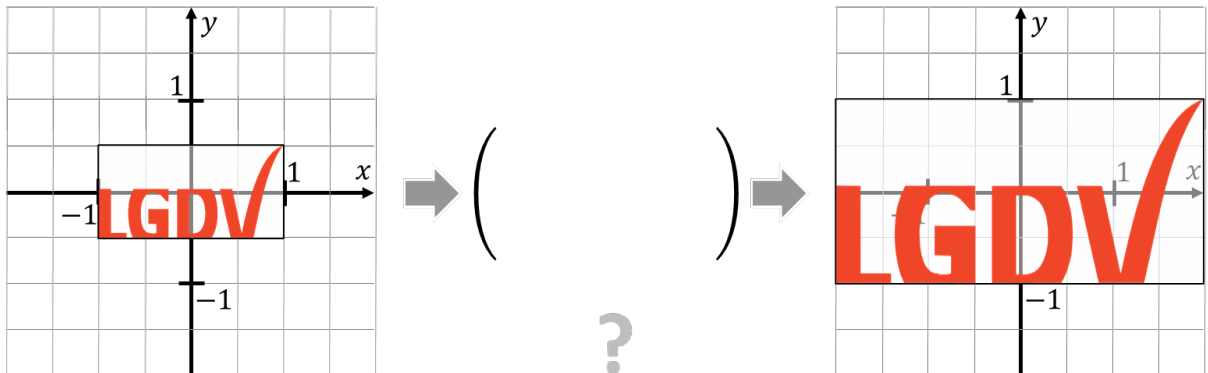
Assignment 1: Affine and Projective Transformations [12 points]

Affine mappings in **2D** can be expressed using **homogeneous coordinates** and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and $\vec{t}$ is the translation:

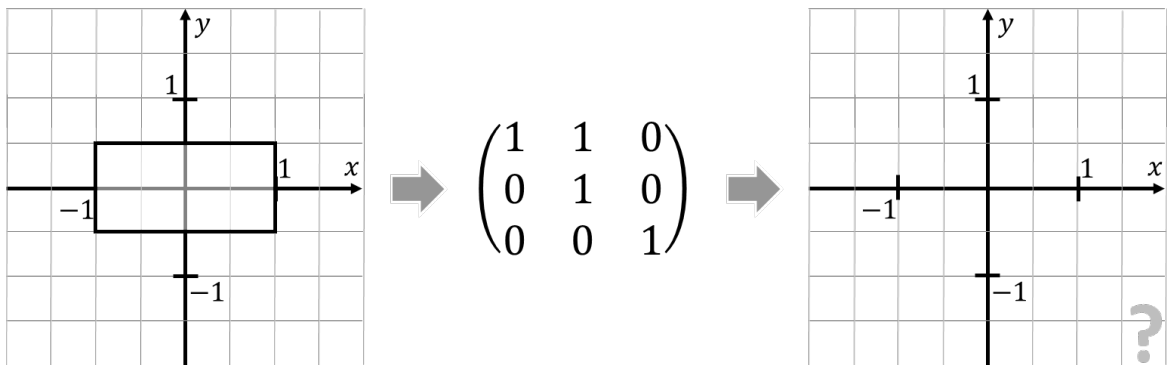
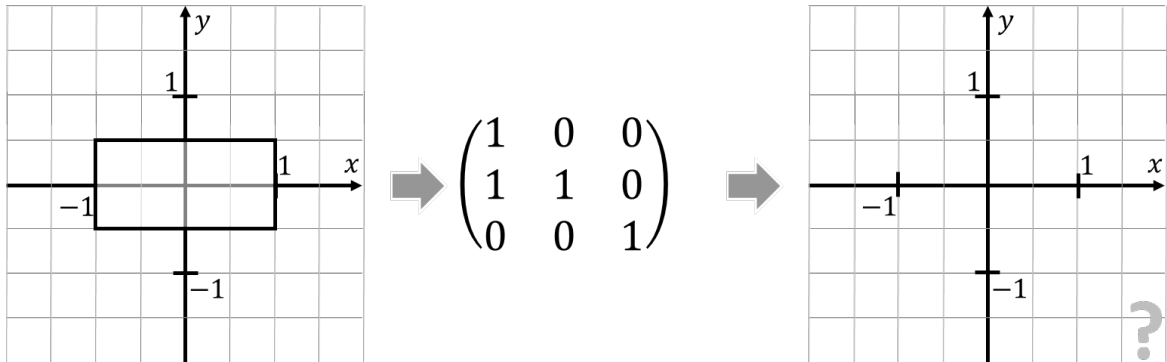


The following illustrations show a shape, which is transformed using an affine matrix.

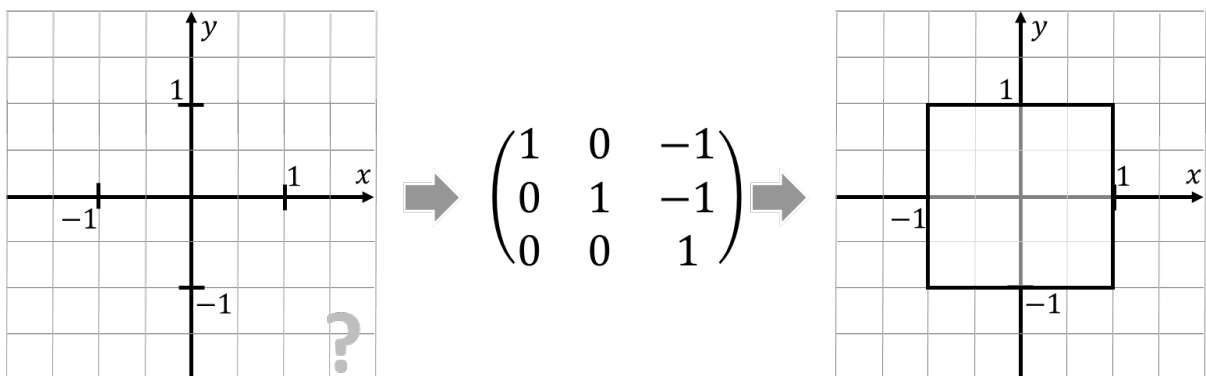
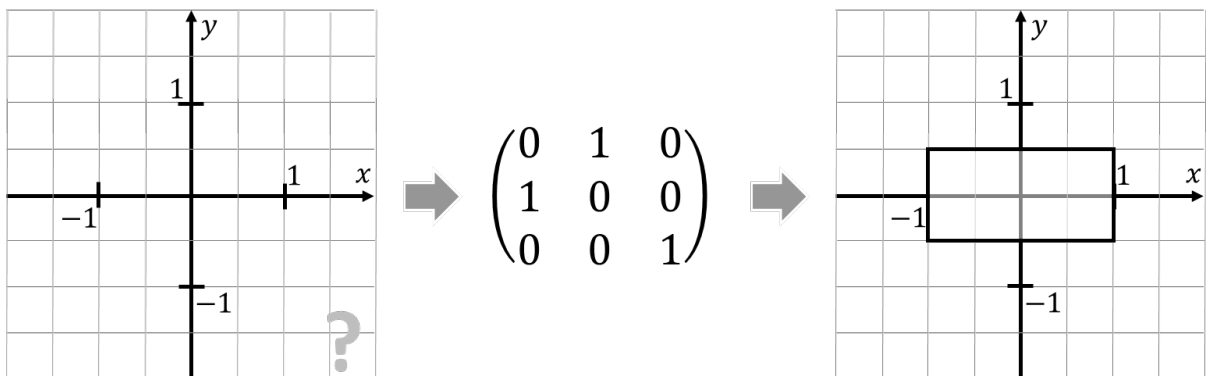
a) [3P] Fill in the matrices for the following mappings



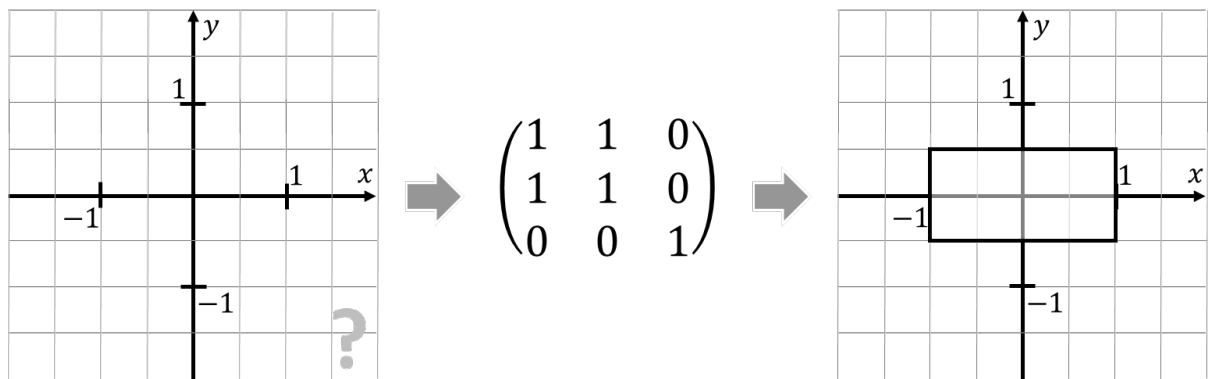
b) [2P] Sketch the results of the following mappings



c) [2P] Sketch the source shapes of the following mappings:



d) [1P] Why does the following problem have no solution?



e) [4P] For the following matrices, add “yes” or “no” for the given properties.

Matrix	Linear	Affine	Rigid	Projective
$\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$				
$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$				
$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 1 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$				
$\begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 1 & 1 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$				

Assignment 2: Perspective [9 points]

a) **[4P]** Given the following perspective matrix:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

which has $(0,0,0)$ as eye point, $(0,0,1)$ as view direction, and $(0,1,0)$ as up-vector.

Also given is a square with vertices $A = (-1, -1, 1)$, $B = (-1, -1, 3)$, $C = (1, -1, 3)$, $D = (1, -1, 1)$.

Map the vertices A, B, C, D with P to screen space and fill in the following table:

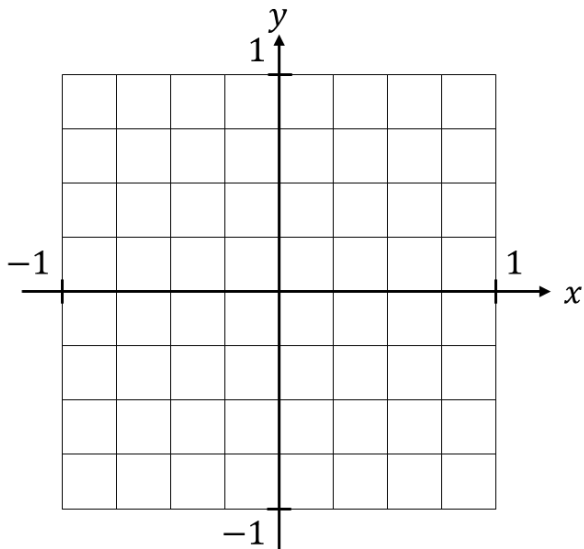
Fill in the following table

Vertex	Screen Space Position (x,y)	Depth (after perspective)
$A = (-1, -1, 1)$		
$B = (-1, -1, 3)$		
$C = (1, -1, 3)$		
$D = (1, -1, 1)$		

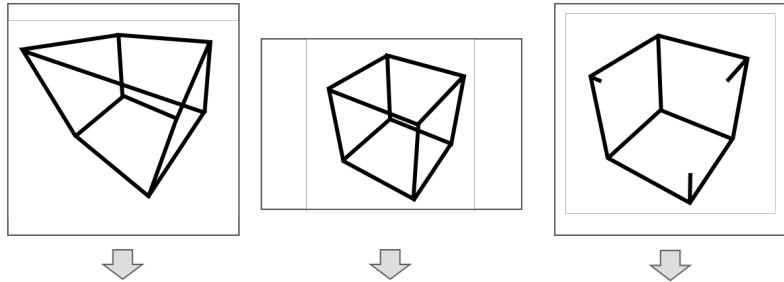
b) [2P] Determine the near plane (z – coordinate after perspective mapping == -1)

$$z_{near} =$$

Sketch the result of the above perspective mapping of the given square, **including clipping through the near plane**



c) **[3P]** Given are the following images of a cube with different camera parameters *fovy* (field of view in *y*-direction), *aspect* (ratio of image width over height), and *near* (distance of near plane). Assign the correct parameter set to the images.



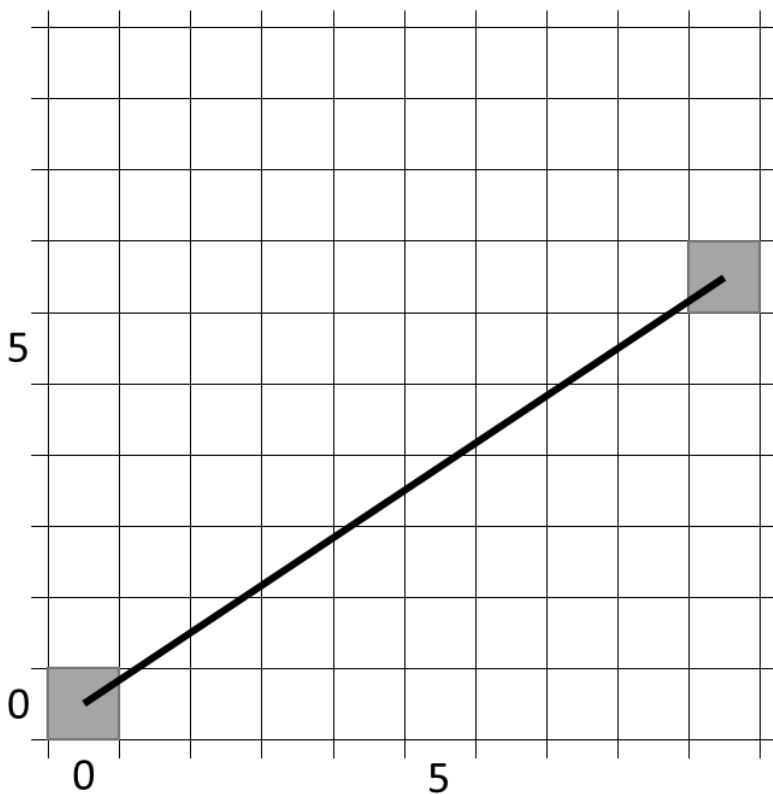
parameter	possible values	image 1	image 2	image 3
field of view	45° or 120°			
aspect ratio	1 or 1.5			
near	1 or 5			
far	100	100	100	100

Assignment 3: Line Rasterization [10 points]

Given the direct line drawing algorithm for lines with $x_1 > x_0$ and $\text{slope} < 1$:
(**round()** rounds to the closest integer)

```
function line(int x0, int y0, int x1, int y1)
{
    float m = (y1 - y0) / (x1 - x0);
    float y = y0;
    for (float x = x0; x <= x1; x++) {
        setPixel(x, round(y));
        y += m;
    }
}
```

a) **[3P]** Rasterize the line (0,0) to (9,6) using this algorithm. Fill in the table on the right and mark the set pixel in the pixel grid below.



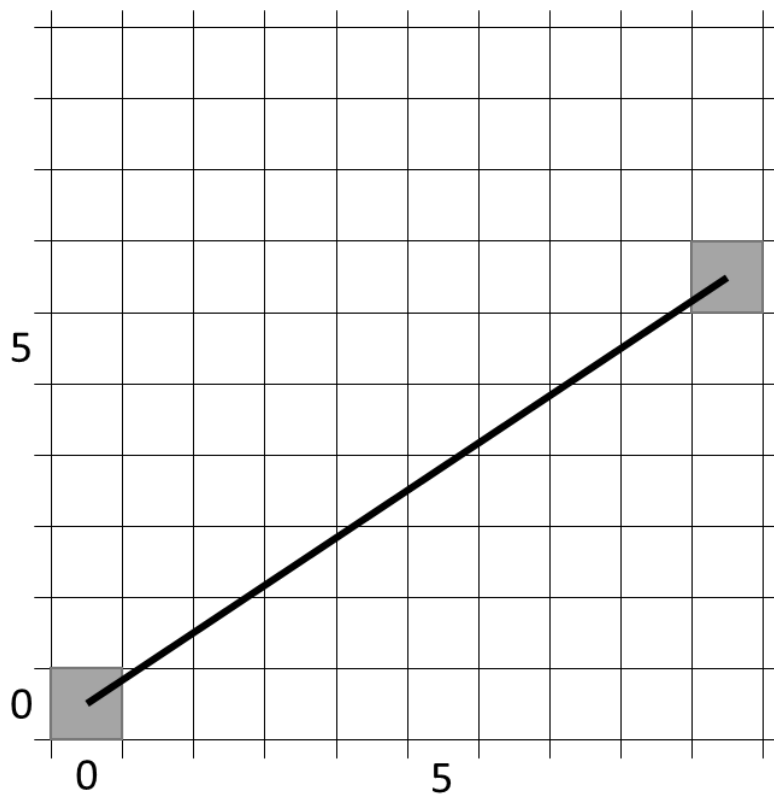
m	2/3
---	-----

x	y	round(y)
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		

We next consider Bresenham's algorithm for drawing lines with $x_1 > x_0$ and $\text{slope} < 1$:

```
function line(int x0, int y0, int x1, int y1)
{
    int x = x0
    int y = y0
    int Δx = x1 - x0
    int Δy = y1 - y0
    int D = Δx - 2Δy
    int ΔDE = -2Δy, ΔDNE = 2(Δx - Δy)
    while (x <= x1)
        setPixel(x,y)
        x = x + 1
        if (D < 0) {
            y = y + 1
            D = D + ΔDNE
        }
        else
```

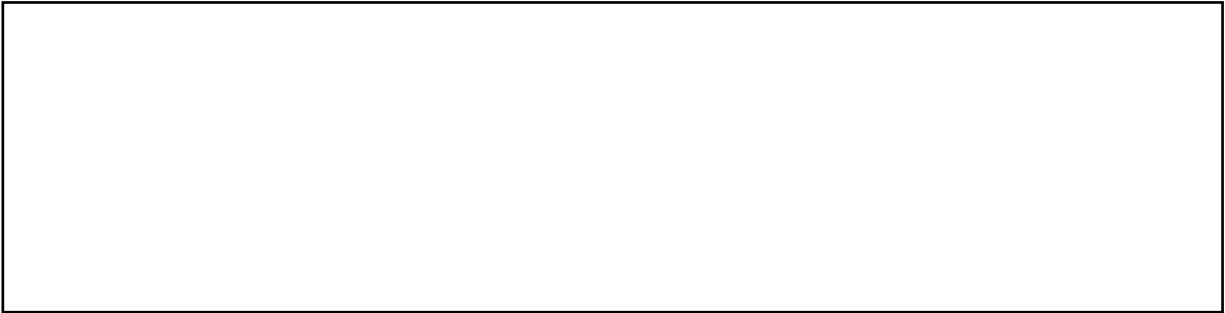
b) **[3P]** Rasterize the same line as in a) from (0,0) to (9,6) with Bresenham's algorithm. Fill in the table on the right and mark the set pixels in the pixel grid below.



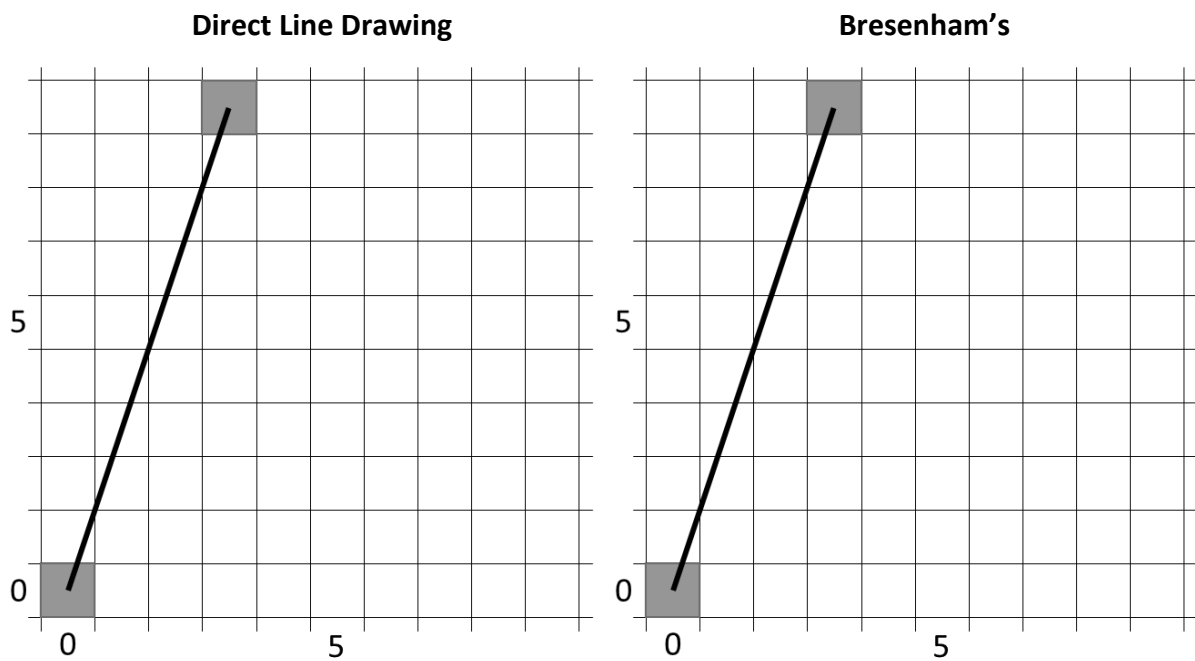
Δx	9
Δy	6
D	
ΔDE	
ΔDNE	

x	D	y
0	-3	0
1		
2		
3		
4		
5		
6		
7		
8		
9		

c) **[2P]** Prove that at the end D has same value as at the beginning (hint: you go Δy times north east and $\Delta x - \Delta y$ times east).

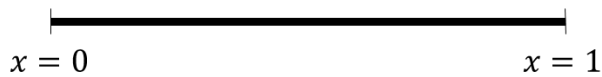


d) **[2P]** Both algorithms assume that $y_1 - y_0 \leq x_1 - x_0$. What happens, if this assumption is not true? Rasterize the line (0,0) to (6,9) using both algorithms. Just mark the set pixels, you don't need to provide intermediate variable values.



Assignment 4: (Blinn-)Phong Lighting & Shading [11 points]

a) **[4P]** Given is the following setup of a camera, a light source, and a surface (line at bottom, parameterized from $x = 0$ to 1).

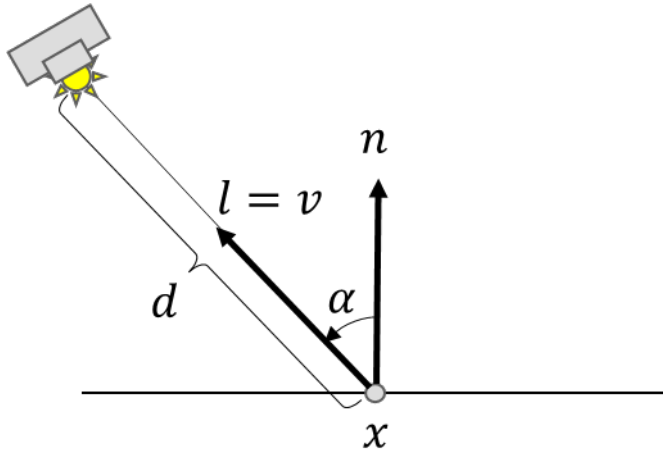


Sketch below the ambient, diffuse, and specular reflection (for different exponents), assuming reflection coefficients $k_{amb} = 0.5$, $k_{diff} = 0.5$, $k_{spec} = 1$, and no distance attenuation of the light source.





b) **[3P]** Let's consider a light source fixated to the camera, e.g. a flash light at a camera, or a miner's lamp. As approximation, we say that the light source is identical to the camera position. We illuminate (and look at) a surface point x with normal n under an incident angle α . The distance from the light source to x is d :

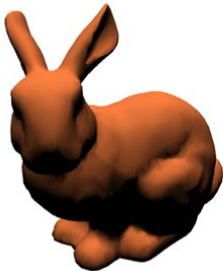
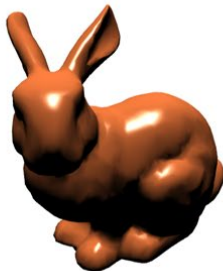
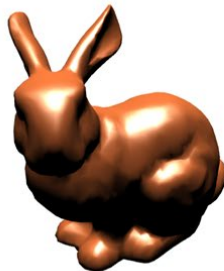


Write down formulas for the diffuse and specular term of Phong lighting including quadratic attenuation w.r.t. distance d . Use a diffuse reflection coefficient k_{diff} , a specular reflection coefficient k_{spec} , the angle α , and the distance d :

c) [4P] Given are following Phong material parameters:

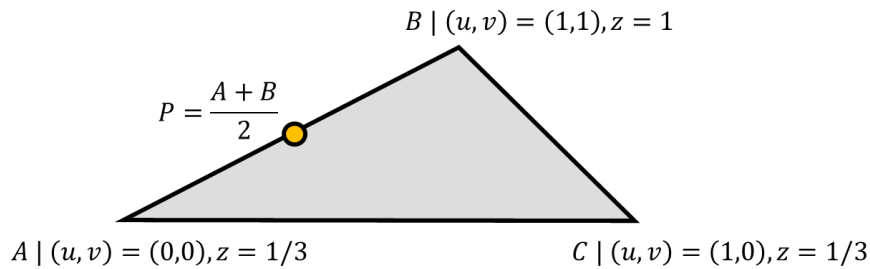
Material Ambient Color  Diffuse Color  Specular Color  nshiny  8	Material Ambient Color  Diffuse Color  Specular Color  nshiny  8	Material Ambient Color  Diffuse Color  Specular Color  nshiny  8
material A	material B	material C
Material Ambient Color  Diffuse Color  Specular Color  nshiny  33	Material Ambient Color  Diffuse Color  Specular Color  nshiny  8	Material Ambient Color  Diffuse Color  Specular Color  nshiny  8
material D	material E	material F

Each of the following images has been rendered using one of the above materials. Assign the right material to each of the following renderings:

			
material	material	material	material

Assignment 5: Texture Mapping [12 points]

Given is a 2D triangle A, B, C in screen space (after perspective projection). Each vertex has texture coordinates (u, v) , plus a z -value from the original 3D triangle.



a) **[1P]** P is the midpoint of A and B *after* perspective projection. What are the barycentric coordinates of P w.r.t. A, B, C in screen space?

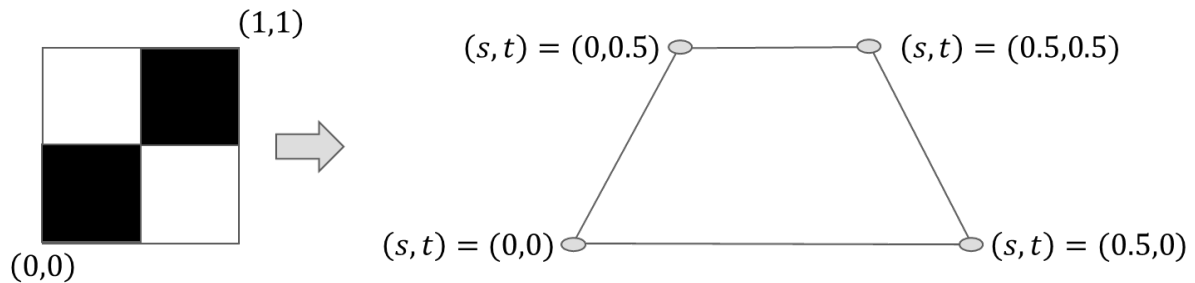
$(\quad , \quad , \quad)$

b) **[4P]** Perform a perspective correct texture coordinate interpolation for point P and write down the interpolated texture coordinates for P :

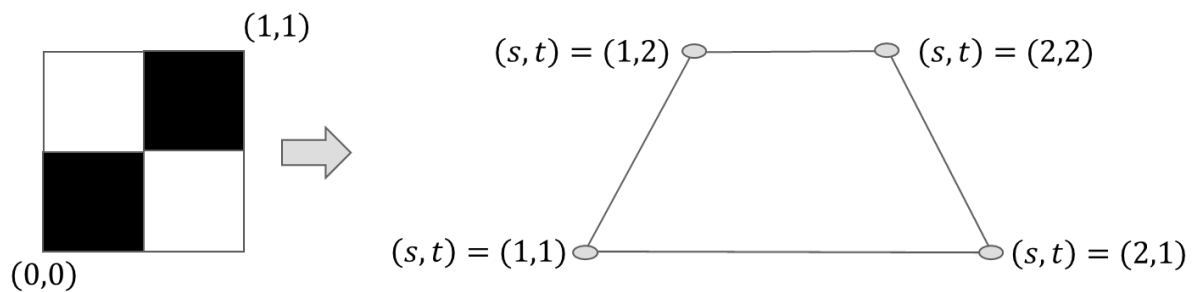
$$(u, v) = (\quad , \quad)$$

c) **[3P]** Sketch the texture mapping result for the left 2x2 texture onto the perspective square on the right, using the provided texture coordinates (s, t) :

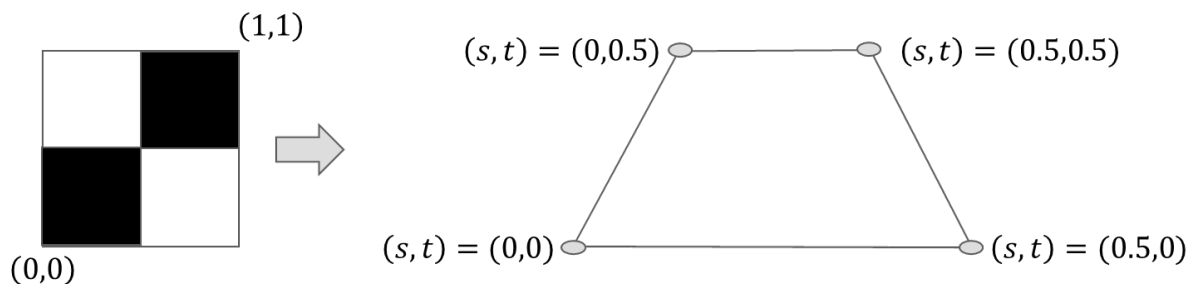
- nearest neighbor interpolation, no MIP-mapping



- nearest neighbor interpolation, no MIP-mapping, texture wrapping mode: REPEAT



- nearest neighbor interpolation, MIP-mapping, **coarsest level**



Next, let's assume the following image is a rendering, where the FAU-Logo is a square texture with 256x256 pixels. The texture is rendered with MIP-mapping. At the shown distance to the building of 60m, MIP-mapping selects exactly the resolution of 64x64 pixels.



c) **[2P]** Fill in the following table showing at which distance which texture resolution is selected by MIP-mapping:

Distance			60m			...
MIP-map resolution			64x64	32x32	16x16	...

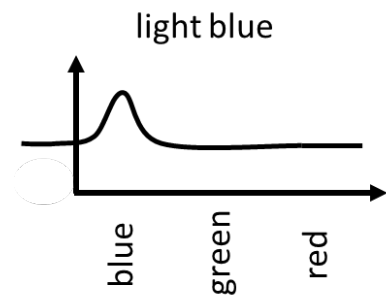
d) **[1P]** At which distance range can we expect aliasing from magnification?

e) **[1P]** For which of the following perspective projections anisotropic MIP-mapping would be most helpful?

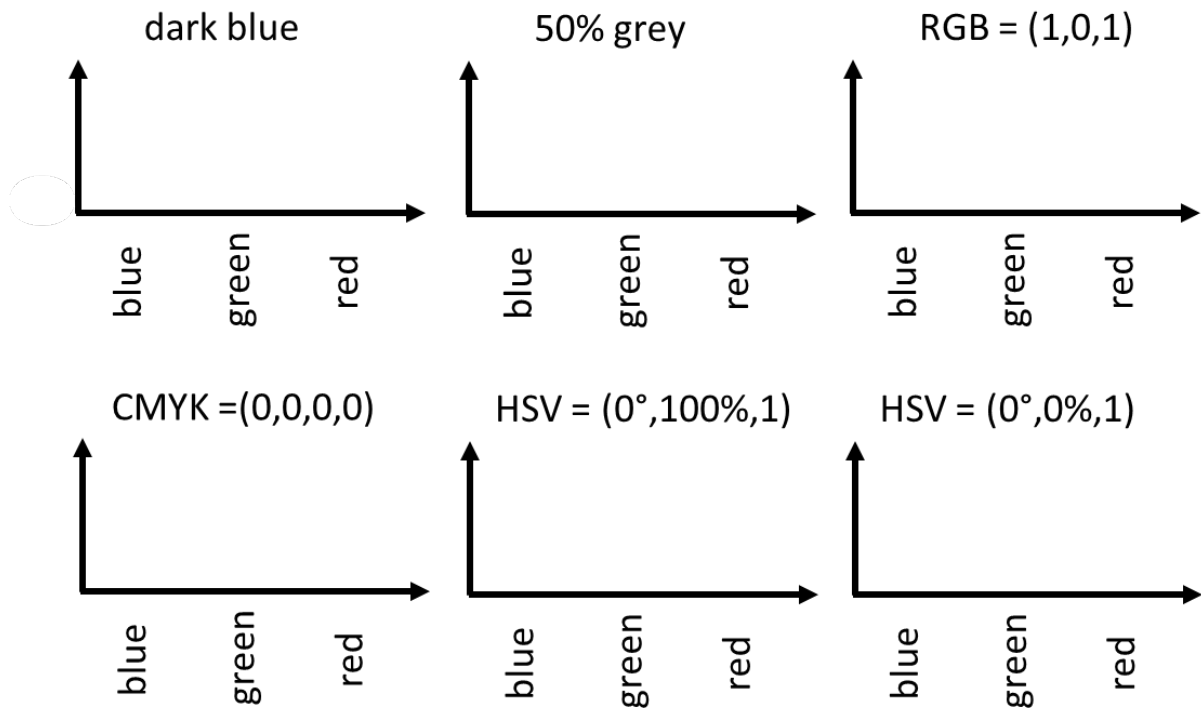


6 Colors [6 points]

Colors can be characterized by their spectrum, a function of wavelengths the visible color is made up of. The visible spectrum roughly goes from blue to left (if the x-axis corresponds to wave length). On the right we can see a spectrum for the color “light blue”



Sketch for the following colors a possible spectrum, where the colors are provided either literally or in one of the color spaces RGB, CMYK, or HSV:



Hint for HSV: 0° corresponds to red

